

2 LITERATURE REVIEW

2.1 Theoretical Background: Endogenous Growth and the Role of Technology

The theoretical foundations for analysing the relationship between technological change and labour productivity are rooted in endogenous growth theory. Romer (1990) formalised the idea that technological change arises from intentional, profit-motivated investment in research, and that the total stock of human capital is the primary determinant of an economy's long-run growth rate. Because knowledge is non-rival, its benefits can be used simultaneously across the economy, creating increasing returns that sustain growth over time. Aghion and Howitt (1992) extended this framework by modelling growth through vertical innovations in which successive technologies render previous ones obsolete — a process of creative destruction. Their model shows that growth depends positively on the magnitude of innovations and on the supply of skilled labour, while warning that market equilibria may deviate from social optima due to intertemporal externalities. Together, these contributions establish that the ability of an economy to convert technological investment into sustained productivity gains is not automatic: it depends critically on the quality of its human capital base and on the institutional environment that governs knowledge creation and diffusion.

2.2 The Modern Productivity Paradox

The central empirical puzzle motivating this study is the so-called productivity paradox: the observation that periods of intense technological innovation do not necessarily translate into commensurate gains in measured aggregate productivity. The phenomenon was coined by Solow (1987), whose celebrated remark that “you can see the computer age everywhere except in the productivity statistics” captured the disconnect between the rapid diffusion of information technology in the 1980s and the concurrent stagnation of measured output per worker. The paradox has gained renewed urgency with the diffusion of artificial intelligence (AI), cloud computing, and digital platforms, which have generated similarly high expectations without an immediate proportional response in aggregate productivity statistics.

Brynjolfsson, Rock and Syverson (2017) provide the most influential contemporary treatment of this paradox. Analysing the coexistence of spectacular AI capabilities with a decade-long deceleration in measured productivity growth across OECD economies, the

authors evaluate four competing explanations: false expectations about the transformative power of AI; measurement errors that cause true productivity gains to be understated in national accounts; redistribution, whereby gains accrue narrowly to capital or specific sectors rather than raising aggregate output; and implementation lags, whereby the full benefits of a general-purpose technology (GPT) materialise only after a lengthy period of complementary reorganisation. The authors conclude that implementation lags are the dominant factor. The most capable AI systems have not yet diffused broadly, and realising their productivity potential requires substantial co-investments in organisational redesign, managerial capabilities, and worker skills — none of which are instantaneous. This analysis implies that the absence of measured productivity gains is not evidence of technological failure but rather a predictable feature of GPT adoption cycles.

Siqueira and Mazali (2026) situate this paradox within the broader context of digital transformation, noting that unlike the steam revolution — which took roughly a century to reshape global production — the digital revolution is restructuring all sectors of the economy simultaneously. Their review highlights that the aggregate productivity slowdown is partly explained by a widening productivity gap between frontier firms, which adopt and exploit digital technologies effectively, and laggard firms, which do not. This inter-firm divergence means that aggregate statistics can mask substantial productivity heterogeneity, and that policies targeting technology diffusion are as important as those promoting technological frontier advancement.

2.3 Automation, Task Displacement, and Labour Market Adjustment

Acemoglu and Restrepo (2018) provide the theoretical framework most directly relevant to understanding how automation affects labour productivity at the macroeconomic level. Their task-based model decomposes the aggregate effect of automation into two opposing forces. The displacement effect captures the substitution of machines for workers in tasks previously performed by labour, tending to reduce labour demand, wages, and the labour share of income. The productivity effect captures the cost savings generated by automation, which reduce production costs and thereby increase demand for labour in tasks that remain human-intensive. Whether automation raises or reduces aggregate labour productivity depends on the relative magnitude of these effects, and crucially on whether automation is accompanied by the creation of new labour-intensive tasks that reintegrate displaced workers into productive employment.

The authors' framework identifies the mismatch between the skill requirements of new technologies and the capabilities of the existing workforce as a key structural constraint on productivity gains from digitalisation. When workers lack the qualifications to perform newly created tasks, the productivity effect is attenuated and the displacement effect dominates, resulting in stagnant or declining labour productivity despite rising technological investment. This mechanism is directly applicable to the Brazilian context, where significant skill heterogeneity and regional disparities in educational attainment may limit the economy's capacity to absorb and productively deploy digital technologies.

Korinek, Schindler and Stiglitz (2021) extend this analysis to the international dimension, arguing that AI and automation technologies are inherently labour-saving and tend to generate winner-takes-all dynamics that favour technologically advanced economies and highly skilled workers. For emerging economies whose comparative advantage rests on abundant low-cost labour, the authors warn that the current wave of automation risks undermining the export-led manufacturing model that historically served as a pathway out of poverty. They argue that realising more inclusive outcomes from technological progress requires coordinated policy action — both domestically, through investments in education and social protection, and globally, through reforms in intellectual property and technology transfer regimes.

2.4 Firm-Level Evidence: Digitalisation, Complementarities, and Productivity Dispersion

The most comprehensive firm-level empirical evidence on the productivity effects of digital adoption is provided by Gal et al. (2019), who combine technology adoption data at the industry level with productivity data at the firm level across multiple EU countries. Their findings confirm a positive and statistically significant association between digital technology penetration within an industry and firm-level productivity. Three key empirical regularities emerge from their analysis.

First, the productivity gains from digitalisation are concentrated among already-productive firms, implying that digital technologies tend to widen the productivity distribution rather than uniformly raising the floor. This result is consistent with the theoretical prediction

that complementary assets — skilled workers, advanced organisational practices, and intangible capital — are prerequisites for converting digital investment into output gains. Firms that already possess these assets benefit disproportionately from new technologies, while those that lack them see limited returns.

Second, skill shortages substantially attenuate the productivity benefits of digitalisation. Gal et al. (2019) find that the positive effect of digital adoption is significantly weaker in industries or regions where qualified workers are scarce, providing direct empirical support for the complementarity between technology and human capital emphasised by both Romer (1990) and Acemoglu and Restrepo (2018). Third, the productivity gains are particularly pronounced in manufacturing activities and routine-based tasks — precisely the categories most susceptible to task-level automation — while effects in service sectors are more heterogeneous. The policy implication drawn by the authors is that promoting digital adoption in isolation is insufficient; measures to raise skills, improve managerial quality, and facilitate organisational adaptation are equally necessary, particularly for laggard firms.

A further dimension of the empirical literature concerns the long-run sustainability of productivity gains under conditions of pervasive AI adoption. Acemoglu, Kong and Ozdaglar (2026) develop a model demonstrating that while agentic AI improves short-run decision-making by supplying context-specific information, it simultaneously erodes the incentives for humans to accumulate general knowledge. Because contributions to the general stock of knowledge generate positive externalities that individual agents do not internalise, the widespread substitution of AI for human cognitive effort may progressively deplete the collective knowledge base, leading to a state of “knowledge collapse” with adverse long-run consequences for productivity and innovation capacity.

2.5 Synthesis and Research Gap

The literature reviewed in this section converges on several key insights. The productivity paradox — the disconnect between technological investment and measured productivity outcomes — is a well-documented empirical regularity first identified by Solow (1987) and subsequently explained by implementation lags, complementarity failures, and inter-firm heterogeneity (Brynjolfsson et al., 2017; Siqueira & Mazali, 2026; Gal et al., 2019). The benefits of digitalisation are not automatic: they depend on the availability of skilled labour,

organisational adaptability, and institutional quality (Romer, 1990; Acemoglu & Restrepo, 2018; Gal et al., 2019). For emerging economies, additional structural constraints — including skill heterogeneity, digital infrastructure gaps, and exposure to labour-saving automation — may amplify the paradox (Acemoglu & Restrepo, 2018; Korinek et al., 2021).

Despite the richness of this international evidence base, the literature applied specifically to Brazil remains underdeveloped. Most firm-level and sector-level empirical studies focus on OECD or EU economies, and the post-2015 period in Brazil — which combined severe economic contraction with accelerating digital transformation — has received limited systematic econometric attention. The present study addresses this gap by empirically examining the relationship between digital technology adoption and labour productivity dynamics in Brazil from 2010 to 2023, with explicit attention to sectoral heterogeneity and to the structural moderating factors identified in the theoretical and empirical literature reviewed above.

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